

CHAPTER 1: MATHEMATICAL MODELS IN MECHANICS

1.1 Constructing a model

- **Definition / Core Concept:** Mechanics deals with the **motion** and the action of **forces** on objects. Mathematical models are constructions used to **simulate real-life situations** (creating conditions existing in real life to study them).
- **Governing Principle:** Modelling involves simplifying a problem by making one or more **assumptions**. This allows the description of problems using **equations or graphs**.
- **Conceptual and Mechanistic Explanation:**
 - The solution to a mathematical model must be **interpreted** within the context of the original problem.
 - Models may require **refinement** (improvement via small changes) and reconsideration of assumptions.
- **The Mathematical Modelling Process:**
 - Real-world problem.
 - Set up a mathematical model:
 - Identify assumptions.
 - Identify variables.
 - Solve and interpret.
 - Check if the answer is **reasonable**:
 - If **Yes**: Report solution.
 - If **No**: Reconsider assumptions and return to step 2.
- [Diagram: Flow chart of the mathematical modelling process showing the loop between "Is your answer reasonable?", "Reconsider assumptions", and "Set up a mathematical model"].
- **Special Cases and Boundary Scenarios:**
 - **Validity Conditions:** Models are often valid only within specific ranges.
 - **Example (Basketball motion):** $h = 2 + 1.1x - 0.1x^2$.
 - Where h is height above ground and x is horizontal distance.
 - **Constraint:** Height h cannot be negative ($h \geq 0$); the model is invalid if the calculation yields a negative height (e.g., when $x = 15$ m, $h = -4$ m, which is invalid).
- **Practical Significance:** Allows for the prediction of behavior in complex physical systems by reducing them to manageable mathematical relations.

1.2 Modelling assumptions

- **Definition:** Simplifications applied to real-life situations to enable analysis using known mathematical techniques.
- **Governing Principle:** The significance of each assumption must be understood as it directly affects the validity of calculations.
- **Standard Models and Corresponding Assumptions:**
 - **Particle:**
 - Dimensions of the object are **negligible**.
 - Mass is concentrated at a **single point**.
 - **Rotational forces** (moving around a central fixed point) and **air resistance** can be ignored.
 - **Rod:**
 - All dimensions except one are negligible (e.g., a pole or beam).
 - Mass is concentrated **along a line**.
 - Has **no thickness**.
 - **Rigid:** Does not bend or buckle.
 - **Lamina:**
 - Object with area but negligible thickness (e.g., sheet of paper).
 - Mass is distributed across a **flat surface**.
 - **Uniform body:**
 - Mass is **distributed evenly**.
 - Mass is concentrated at a single point at the geometric centre—the **centre of mass**.
 - **Light object:**
 - Mass is small compared to other masses (e.g., string or pulley).
 - Treated as having **zero mass**.
 - **Tension** is the same at both ends of a light string.
 - **Inextensible string:**
 - A string that does not stretch under load.
 - **Acceleration** is the same in objects connected by a taut inextensible string.
 - **Smooth surface:**
 - Assume there is **no friction** between the surface and any object on it.
 - **Rough surface:**
 - Objects in contact experience a **frictional force** if they are moving or acted on by a force.
 - **Wire:**
 - Rigid thin length of metal.
 - Treated as **one-dimensional**.
 - **Smooth and light pulley:**
 - Pulley has **no mass**.
 - **Tension** is the same on either side of the pulley.
 - **Bead:**
 - Particle with a hole for threading on a wire or string.
 - Moves **freely** along a wire or string.

- Tension is the same on either side of the bead.
 - **Peg:**
 - A support from which a body can be suspended or rested.
 - **Dimensionless** and **fixed**.
 - Can be rough or smooth as specified.
 - **Air resistance:**
 - Resistance experienced as an object moves through air.
 - Usually modelled as being **negligible**.
 - **Gravity:**
 - Force of attraction between all objects.
 - Assume all objects with mass are attracted toward the Earth.
 - Earth's gravity is **uniform**: acceleration is the same in all parts, at all times, and acts **vertically downward**.
 - Acceleration due to gravity is denoted by g .
 - **Standard Constant**: $g = 9.8 \text{ m s}^{-2}$ (unless otherwise stated).
- **Limitations:** Modelling assumptions can compromise accuracy (e.g., ignoring wind/air resistance for an aeroplane flight is inappropriate).

1.3 Quantities and units

- **Core Concept:** Mechanics uses the International System of Units (**SI units**).
- **Base SI Units:**
 - **Mass:** kilogram (kg).
 - **Length/Displacement:** metre (m).
 - **Time:** second (s).
- **Derived SI Units:**
 - **Speed/Velocity:** metres per second ($m \text{ s}^{-1}$).
 - **Acceleration:** metres per second per second ($m \text{ s}^{-2}$).
 - **Weight/Force:** newton (N).
 - Unit equivalence: $1 \text{ N} = 1 \text{ kg m s}^{-2}$.
- **Critical Distinction:** Kilograms measure **mass**; Newtons measure **weight** (a force).
- **Common Forces in Mechanics:**
 - **Weight:** Gravitational force acting **vertically downward**.
 - **Normal reaction:** Force acting **perpendicular** (90° angle) to a surface when an object is in contact with it.
 - **Friction:** Force opposing motion between two **rough surfaces**.
 - **Tension:** Force acting on an object being **pulled** along a string.
 - **Thrust / Compression:** Force acting on an object being **pushed** along a light rod.
 - **Buoyancy:** Upward force on a body allowing it to float/rise when submerged in liquid.
 - **Air resistance:** Force opposing motion through air.
 - **Lift:** Upward force keeping an aircraft in the air.

- **Forward thrust:** Force propelling an aircraft forward.
 - **Drag:** Force acting in the opposite direction to forward thrust (often synonymous with air resistance in aircraft).
 - **Force Diagrams:**
 - [Diagram: Book resting on a table. Labels: R (Normal reaction) upward, W (Weight) downward. Arrows indicate forces have the same magnitude].
 - [Diagram: Object being pulled by a string. Labels: Direction of motion, Friction force (opposing), Tension in string (pulling)].
 - [Diagram: Object being pushed by a rod. Labels: Thrust or compression in rod].
 - [Diagram: Boat in water. Labels: Buoyancy (upward), Weight (downward)].
 - [Diagram: Parachutist. Labels: Air resistance (upward), Weight (downward)].
 - [Diagram: Aircraft in flight. Labels: A (Upward thrust/lift), B (Forward thrust), C (Weight), D (Air resistance/drag)].
 - **Conversion Factors:**
 - $1 \text{ km} = 1000 \text{ m}$.
 - $1 \text{ kg} = 1000 \text{ g}$.
 - Convert km h^{-1} to m s^{-1} : Multiply by 1000 and divide by 3600 (60×60).
 - **Significance:** Standardization of units ensures consistency in mechanical calculations and physical interpretations.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Gravity Constant:** $g = 9.8 \text{ m s}^{-2}$.
 - **Force Unit Definition:** $1 \text{ N} = 1 \text{ kg m s}^{-2}$.
 - **Height Example Relation:** $h = 2 + 1.1x - 0.1x^2$.
 - **Mass Symbol:** kg (kilogram).
 - **Length/Displacement Symbol:** m (metre).
 - **Time Symbol:** s (second).
 - **Velocity Symbol:** m s^{-1} .
 - **Acceleration Symbol:** m s^{-2} .
 - **Force Symbol:** N (newton).
 - **Normal Reaction Symbol:** R .
 - **Weight Symbol:** W .
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ULTRA-CONDENSED FINAL SUMMARY

- **Mechanics Foundation:** Study of motion and forces using mathematical models.
- **The Process:** Define problem -> Assume/Simplify -> Solve -> Validate -> Refine.
- **Core Assumptions:**
 - **Particles** have mass but no size; ignore rotation/air resistance.
 - **Light** objects have zero mass; **Inextensible** strings have constant length and acceleration.
 - **Smooth** means zero friction; **Uniform** means mass is at the geometric centre.
 - **Gravity** (g) is constant (9.8 m s^{-2}) and acts vertically downward.
- **Units:** Always use SI (kg, m, s, N). Distinguish mass (kg) from weight (N).
- **Forces:** Weight (down), Normal Reaction (perpendicular), Friction (opposes motion), Tension (pull), Thrust (push).

CHAPTER 2: CONSTANT ACCELERATION

2.1 Displacement–time graphs

- **Definition / Core Concept:** A graphical representation of an object's motion where displacement (s) is plotted on the vertical axis and time (t) is plotted on the horizontal axis.
- **Governing Principle:**
 - The **gradient** of a displacement–time graph represents the **velocity** (v).
 - If the graph is a straight line, the velocity is **constant**.
- **Conceptual and Mechanistic Explanation:**
 - **Stationary Object:** Represented by a horizontal line (gradient is zero, so velocity is zero).
 - **Constant Velocity:** Represented by a straight line with a constant slope; the object covers equal displacement in equal time intervals.
 - **Accelerating Object:** Represented by a curve where the gradient increases over time (steeper slope means higher velocity).
- **Equations and Derived Forms:**
 - Average velocity = $\frac{\text{displacement from starting point}}{\text{time taken}}$.
 - Average speed = $\frac{\text{total distance travelled}}{\text{time taken}}$.
- **Special Cases and Boundary Scenarios:**

- **Negative Velocity:** If the graph has a negative gradient, the object is moving back toward the starting point (displacement is decreasing).
- **Zero Average Velocity:** Occurs if the object returns to its exact starting position, regardless of distance travelled.
- [Diagram: Three graphs showing: 1. Horizontal line (stationary), 2. Straight sloping line (constant velocity), 3. Upward curve (increasing velocity/acceleration)].
- [Diagram: Displacement-time graph with labels $s(m)$ on y-axis and $t(s)$ on x-axis].

2.2 Velocity–time graphs

- **Definition / Core Concept:** A graphical representation of motion where velocity (v) is plotted on the vertical axis and time (t) is on the horizontal axis.
- **Governing Principle:**
 - The **gradient** represents the **acceleration** (a).
 - The **area between the graph and the horizontal axis** represents the **displacement** (s).
- **Conceptual and Mechanistic Explanation:**
 - **Stationary:** Velocity is zero (line stays on the t -axis).
 - **Constant Velocity:** Represented by a horizontal line (gradient is zero, so acceleration is zero).
 - **Constant Acceleration:** Represented by a straight line with a constant gradient.
 - **Deceleration (Retardation):** Represented by a negative gradient; the object is slowing down.
- **Equations and Derived Forms:**
 - For motion in a straight line with constant velocity $V \text{ m s}^{-1}$ for time T seconds:
 - Displacement = velocity $\times$ time.
 - Area under graph = $V \times T$.
 - For constant acceleration (trapezium area):
 - $s = \frac{1}{2}(a + b)h$ where a and b are parallel vertical sides (velocities) and h is time.
- **Practical Significance:** Allows simultaneous calculation of acceleration and distance travelled from a single visual model.
- [Diagram: Velocity-time graph showing a horizontal line with shaded area underneath labeled "Area under the graph = $V \times T$ ".]
- [Diagram: Velocity-time graph showing a straight line with a positive slope (constant acceleration) and a negative slope (deceleration)].

2.3 Acceleration–time graphs

- **Definition / Core Concept:** A graph showing the rate of change of velocity over time. Acceleration (a) is on the vertical axis and time (t) is on the horizontal axis.

- **Governing Principle:** The **area between the acceleration–time graph and the horizontal axis** represents the **change in velocity**.
- **Conceptual and Mechanistic Explanation:**
 - **Zero Acceleration:** Horizontal line at $a = 0$ (constant velocity).
 - **Positive Acceleration:** Horizontal line above the t -axis (velocity is increasing).
 - **Negative Acceleration:** Horizontal line below the t -axis (velocity is decreasing).
- **Assumptions and Limitations:**
 - This syllabus only considers constant acceleration; therefore, acceleration–time graphs will consist of horizontal line segments only.
 - Sloping lines on these graphs (changing acceleration) are beyond the scope.
- **Equations:**
 - Change in velocity = $a \times t$ (area of the rectangle).
 - Final velocity = Initial velocity + Change in velocity.
- [Diagram: Acceleration-time graphs showing horizontal lines for zero, positive, and negative acceleration].

2.4 Constant acceleration formulae 1

- **Definition / Core Concept:** Derivation of the first two fundamental equations for objects moving in a straight line with constant acceleration.
- **Standard Symbols:**
 - s : displacement (m).
 - u : initial velocity ($m\ s^{-1}$).
 - v : final velocity ($m\ s^{-1}$).
 - a : acceleration ($m\ s^{-2}$).
 - t : time (s).
- **Derivations:**
 - **Equation 1 (Velocity-Time relation):**
 - From gradient of $v - t$ graph: $a = \frac{v - u}{t}$.
 - Rearranging: $v = u + at$.
 - **Equation 2 (Displacement-Time relation):**
 - From area of trapezium under $v - t$ graph: $s = \text{average velocity} \times \text{time}$.
 - Formula: $s = \left(\frac{u + v}{2} \right) t$.
- **Assumptions / Constraints:** Acceleration **must** be constant for these formulae to be valid.
- [Diagram: Velocity-time graph from $t = 0$ to $t = t$ with initial velocity u and final velocity v . The line has gradient a and the shaded area is the displacement s].

2.5 Constant acceleration formulae 2

- **Definition:** Derivation of the remaining three "suvat" or "kinematics" formulae by eliminating variables.
- **Derivations:**
 - **Equation 3 (Eliminating t):**
 - From (1): $t = \frac{v - u}{a}$.
 - Substitute into (2): $s = \left(\frac{u + v}{2}\right) \left(\frac{v - u}{a}\right)$.
 - Result: $v^2 = u^2 + 2as$.
 - **Equation 4 (Eliminating v):**
 - Substitute (1) $v = u + at$ into (2): $s = \left(\frac{u + u + at}{2}\right) t$.
 - Result: $s = ut + \frac{1}{2}at^2$.
 - **Equation 5 (Eliminating u):**
 - Substitute $u = v - at$ into (4): $s = (v - at)t + \frac{1}{2}at^2$.
 - Result: $s = vt - \frac{1}{2}at^2$.
- **Summary List of Kinematics (suvat) Formulae:**
 - $v = u + at$
 - $s = \left(\frac{u + v}{2}\right) t$
 - $v^2 = u^2 + 2as$
 - $s = ut + \frac{1}{2}at^2$
 - $s = vt - \frac{1}{2}at^2$
- **Special Cases:**
 - **Return to start:** Displacement $s = 0$.
 - **Change of direction:** Velocity $v = 0$ momentarily at the furthest point.
 - **Distance vs Displacement:** Total distance travelled is the sum of magnitudes of individual path segments (e.g., if an object moves 45 m out and returns, total distance is 90 m while displacement is 0 m).

2.6 Vertical motion under gravity

- **Definition / Core Concept:** Application of constant acceleration formulae to objects moving vertically under the influence of Earth's gravitational pull.
- **Governing Principle:** Gravity causes all objects to accelerate toward Earth at a constant rate, regardless of mass (in a vacuum).
- **Assumptions and Conditions:**

- Air resistance is ignored (negligible).
 - The object is modeled as a **particle**.
 - Acceleration due to gravity is constant: $g = 9.8 \text{ m s}^{-2}$.
 - **Sign Convention and Mechanistic Reasoning:**
 - You must choose a positive direction (upward or downward).
 - **Upward as Positive:** Acceleration $a = -9.8 \text{ m s}^{-2}$ (because gravity acts downward).
 - **Downward as Positive:** Acceleration $a = +9.8 \text{ m s}^{-2}$.
 - **Special Cases and Boundary Conditions:**
 - **Maximum Height:** The point where vertical velocity becomes zero ($v = 0$).
 - **Time of Flight:** Total time from projection until the object hits the ground/original level.
 - **Speed of Projection:** The initial velocity u at the start of motion.
 - **Symmetry:** For an object projected and returning to the same level, time to reach max height equals time to fall back.
 - **Technical Rigor (Accuracy):** Final answers should be consistent with the accuracy of g (usually 2 significant figures if $g = 9.8$ is used).
 - [Diagram: Feather and apple falling at the same rate in a vacuum with labels $g = 9.8 \text{ m s}^{-2}$].
 - [Diagram: Ball projected upward from point X at 7 m height. Labels: Max height reached, trajectory showing upward and downward path].
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Displacement:** $s \text{ (m)}$
- **Initial Velocity:** $u \text{ (m s}^{-1}\text{)}$
- **Final Velocity:** $v \text{ (m s}^{-1}\text{)}$
- **Acceleration:** $a \text{ (m s}^{-2}\text{)}$
- **Time:** $t \text{ (s)}$
- **Gravity Constant:** $g = 9.8 \text{ m s}^{-2}$
- **Average Velocity:** $\text{Avg } v = \frac{s}{t}$
- **Average Speed:** $\text{Avg speed} = \frac{\text{total dist}}{t}$
- **suvat Equation 1:** $v = u + at$
- **suvat Equation 2:** $s = \frac{(u + v)}{2}t$

- **suvat Equation 3:** $v^2 = u^2 + 2as$
 - **suvat Equation 4:** $s = ut + \frac{1}{2}at^2$
 - **suvat Equation 5:** $s = vt - \frac{1}{2}at^2$
 - **Velocity from gradient:** $v = \frac{ds}{dt}$ (on $s - t$ graph)
 - **Acceleration from gradient:** $a = \frac{dv}{dt}$ (on $v - t$ graph)
 - **Displacement from area:** $s = \int v, dt$ (Area under $v - t$ graph)
 - **Change in velocity from area:** $\Delta v = \int a, dt$ (Area under $a - t$ graph)
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ULTRA-CONDENSED FINAL SUMMARY

- **Motion Graphs:**
 - $s - t$: Gradient = Velocity.
 - $v - t$: Gradient = Acceleration; Area = Displacement.
 - $a - t$: Area = Change in Velocity.
- **SUVAT Requirements:** Straight line motion AND **constant acceleration** only.
- **The 5 Kinematics Formulae:** $v = u + at$, $s = \frac{(u + v)}{2}t$, $v^2 = u^2 + 2as$,
 $s = ut + \frac{1}{2}at^2$, $s = vt - \frac{1}{2}at^2$.
- **Gravity (g):** Acts vertically downward at 9.8 m s^{-2} . Independent of mass. Ignore air resistance.
- **Key Conditions:**
 - Max height $\rightarrow v = 0$.
 - Return to start $\rightarrow s = 0$.
 - At rest $\rightarrow v = 0$.
- **Sign Convention:** Critical in gravity problems. If Up is $+$, then $a = -9.8$. If Down is $+$, then $a = +9.8$.

CHAPTER 3: VECTORS IN MECHANICS

3.1 Working with vectors

- **Definition / Core Concept:**
 - A **vector** is a quantity which has both **magnitude** and **direction**.
 - A **scalar** quantity has **magnitude only**.
- **Governing Law / Principle:**
 - **Vector addition:** When vectors are written in terms of unit vectors, they are added by summing the components separately.
 - **Triangle law of addition:** Represented as $\vec{AC} = \vec{AB} + \vec{BC}$.
- **Conceptual and Mechanistic Explanation:**
 - **Vector Quantities:**
 - **Displacement:** Distance in a particular direction; Unit: m .
 - **Velocity:** Rate of change of displacement; Unit: m s^{-1} .
 - **Acceleration:** Rate of change of velocity; Unit: m s^{-2} .
 - **Force/Weight:** Described by magnitude, direction, and point of application; Unit: N .
 - **Scalar Quantities:**
 - **Distance:** Measure of length; Unit: m .
 - **Speed:** Measure of how quickly a body moves; Unit: m s^{-1} .
 - **Time:** Measure of ongoing events; Unit: s .
 - **Mass:** Quantity of matter; Unit: kg .
- **Assumptions and Conditions:**
 - Scalar quantities are **always positive**.
 - In 1D motion, vector quantities can be positive or negative.
 - **Unit Vectors:** Vectors of length 1. Standard unit vectors along Cartesian axes are denoted by $\mathbf{i}$ and $\mathbf{j}$.
- **Equations and Symbols:**
 - Vector form: $a\mathbf{i} + b\mathbf{j}$.
 - Magnitude (Distance from origin): $OB = \sqrt{OA^2 + AB^2}$.
 - Angle/Bearing: $\tan(\angle AOB) = \frac{\text{opposite}}{\text{adjacent}}$. Bearings are measured clockwise from North.
- [Diagram: Cartesian axes showing unit vectors $\mathbf{i}$ at $(1, 0)$ and $\mathbf{j}$ at $(0, 1)$].
- [Diagram: Triangle of vector addition showing vectors $\vec{AB}$ and $\vec{BC}$ forming resultant $\vec{AC}$].

3.2 Solving problems with vectors written using $\mathbf{i}$ and $\mathbf{j}$ notation

- **Definition / Core Concept:** Manipulation of 2D vectors using component-wise arithmetic and trigonometry.

- **Governing Principle:**
 - **Addition/Subtraction:** $(a\mathbf{i} + b\mathbf{j}) \pm (c\mathbf{i} + d\mathbf{j}) = (a \pm c)\mathbf{i} + (b \pm d)\mathbf{j}$.
 - **Magnitude:** Calculated using Pythagoras' Theorem.
- **Conceptual and Mechanistic Explanation:**
 - Vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular; they form the legs of a right-angled triangle where the vector itself is the hypotenuse.
 - To find the angle θ between a vector $x\mathbf{i} + y\mathbf{j}$ and the positive x -axis: $\tan \theta = \frac{y}{x}$.
- **Special Cases:**
 - **Parallel Vectors:** If $\mathbf{a}$ is parallel to $\mathbf{b}$, then $\mathbf{a} = \lambda\mathbf{b}$ for some scalar λ .
 - **Parallel to $\mathbf{i}$:** The $\mathbf{j}$ component must be 0.
 - **Parallel to $\mathbf{j}$:** The $\mathbf{i}$ component must be 0.
- **Equations:**
 - Magnitude of vector $\mathbf{a}$: $|\mathbf{a}| = \sqrt{x^2 + y^2}$.
- [Diagram: Vector $\mathbf{v} = 3\mathbf{i} - 7\mathbf{j}$ shown as a right-angled triangle with horizontal component 3 and vertical component -7].

3.3 The velocity of a particle as a vector

- **Definition / Core Concept:**
 - **Velocity ($\mathbf{v}$):** A vector in the direction of motion.
 - **Speed:** The magnitude of the velocity vector, $|\mathbf{v}|$.
- **Governing Principle:** For constant velocity, displacement is the product of velocity and time.
- **Conceptual and Mechanistic Explanation:**
 - If a particle moves with constant velocity $\mathbf{v} \text{ m s}^{-1}$, after t seconds its displacement is $\mathbf{v}t \text{ m}$.
 - The displacement is parallel to the velocity vector.
- **Equations:**
 - Displacement: $\mathbf{s} = \mathbf{v}t$.
 - Distance: $\text{Distance} = |\mathbf{s}| = |\mathbf{v}|t$.
- **Practical Significance:** Allows tracking of positions in 2D space over time under steady motion.

3.4 Solving problems involving velocity and time using vectors

- **Definition / Core Concept:** Kinematics of particles in 2D using position, velocity, and acceleration vectors.
- **Governing Principle / Theory:**

- **Position Vector:** If a particle starts at $\mathbf{r}_0$ and moves with constant velocity $\mathbf{v}$, its position $\mathbf{r}$ at time t is $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$.
 - **Acceleration Vector ($\mathbf{a}$):** Describes how velocity changes with time.
 - **Force Vector ($\mathbf{F}$):** $\mathbf{F} = m\mathbf{a}$, where m is mass.
 - **Conceptual and Mechanistic Explanation:**
 - A resultant force causes acceleration in the same direction as the force.
 - **Equilibrium:** If a particle is at rest or moving with constant velocity, the resultant of all forces acting on it is the zero vector $\mathbf{0}$.
 - **Assumptions and Conditions:**
 - Motion is considered in a straight line or 2D plane.
 - Acceleration $\mathbf{a}$ is assumed constant for the validity of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.
 - **Equations and Derivations:**
 - Position: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$.
 - Final Velocity: $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.
 - Force: $\mathbf{F} = m\mathbf{a}$.
 - Displacement: $\mathbf{s} = \mathbf{r} - \mathbf{r}_0 = \mathbf{v}t$ (for constant $\mathbf{v}$).
 - **Special Cases and Boundary Scenarios:**
 - **Collision:** Two particles A and B collide if their position vectors $\mathbf{r}_A$ and $\mathbf{r}_B$ are equal at the same time t .
 - **Intercept:** Particle M intercepts S if they are at the same position at a specific time.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Vector Components:** $a\mathbf{i} + b\mathbf{j}$ (Unit vectors $\mathbf{i}$ and $\mathbf{j}$).
 - **Magnitude of Vector:** $|\mathbf{a}| = \sqrt{x^2 + y^2}$.
 - **Speed:** $\text{Speed} = |\mathbf{v}|$.
 - **Displacement (Constant $\mathbf{v}$):** $\mathbf{s} = \mathbf{v}t$.
 - **Distance:** $\text{Distance} = |\mathbf{s}| = |\mathbf{v}|t$.
 - **Position Vector:** $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$.
 - **Velocity with Constant Acceleration:** $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.
 - **Force (Newton's Second Law):** $\mathbf{F} = m\mathbf{a}$.
 - **Equilibrium Condition:** $\sum \mathbf{F} = \mathbf{0}$.
 - **Bearing Angle:** $\tan \theta = \frac{\text{opp}}{\text{adj}}$.
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ULTRA-CONDENSED FINAL SUMMARY

- **Vector vs Scalar:** Vectors (displacement, velocity, acceleration, force) have magnitude AND direction; Scalars (distance, speed, time, mass) have magnitude only.
- **Unit Vectors:** $\mathbf{i}$ (East/Positive x) and $\mathbf{j}$ (North/Positive y) have length 1.
- **Magnitude/Speed:** Use Pythagoras: $\sqrt{x^2 + y^2}$.
- **Position Vector:** $\mathbf{r} = \text{initial position } (\mathbf{r}_0) + \text{displacement } (\mathbf{v}t)$.
- **Constant Acceleration:** $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ and $\mathbf{F} = m\mathbf{a}$.
- **Parallelism:** $\mathbf{a}$ is parallel to $\mathbf{b}$ if $\mathbf{a} = \lambda\mathbf{b}$.
- **Equilibrium:** Net force vector is $\mathbf{0}$.
- **Bearings:** Always clockwise from North (three-figure notation).

CHAPTER 4: DYNAMICS OF A PARTICLE MOVING IN A STRAIGHT LINE

4.1 Force diagrams

- **Definition / Core Concept:** A force diagram is a visual representation showing all forces acting on an object as arrows pointing in their direction of action.
- **Governing Law / Principle:**
 - **Newton's first law of motion:** An object at rest will stay at rest, and an object moving with constant velocity will continue to move with constant velocity unless an unbalanced force acts on it.
 - **Equilibrium:** When the forces acting upon an object are balanced, the object is in equilibrium.
 - **Resultant Force Principle:** A non-zero resultant force acting on an object causes it to accelerate in the same direction as that resultant force.
- **Conceptual and Mechanistic Explanation:**
 - Forces are resolved in a specific direction to find the resultant.
 - The direction of resolution is chosen as positive; forces in the opposite direction are subtracted.
- **Assumptions and Conditions:**
 - **Constant velocity:** Means neither speed nor direction is changing ($a = 0$).
- **Equations and Symbols:**
 - Resultant force horizontal: $R(\rightarrow)$.
 - Resultant force vertical: $R(\uparrow)$.

- [Diagram: Block on rough horizontal plane pulled by force P ; labels: R (normal reaction), W (weight), F (friction/resistance), P (pulling force)].
- [Diagram: Particle acted on by four perpendicular forces: 30 N up, 10 N down, 20 N left, 20 N right].

4.2 Forces as vectors

- **Definition / Core Concept:** Forces can be represented using $\mathbf{i}$ – $\mathbf{j}$ notation or as column vectors.
- **Governing Law / Principle:**
 - **Vector Resultant:** The resultant of two or more forces given as vectors is found by adding the vectors ($\sum \mathbf{F}$).
 - **Equilibrium:** If a particle is in equilibrium, the resultant vector force is the zero vector, $0\mathbf{i} + 0\mathbf{j}$.
- **Conceptual and Mechanistic Explanation:**
 - Forces in $\mathbf{i}$ and $\mathbf{j}$ components are perpendicular and can be treated independently.
 - Magnitude of resultant $\mathbf{R}$ is calculated via Pythagoras.
 - Direction is found using trigonometry ($\tan \theta$).
- **Assumptions and Conditions:**
 - $\mathbf{i}$ represents unit vector due East (positive x -direction).
 - $\mathbf{j}$ represents unit vector due North (positive y -direction).
- **Equations and Derived Forms:**
 - Resultant: $(p\mathbf{i} + q\mathbf{j}) + (r\mathbf{i} + s\mathbf{j}) = (p + r)\mathbf{i} + (q + s)\mathbf{j}$.
 - Magnitude: $|\mathbf{R}| = \sqrt{x^2 + y^2}$.
 - Bearing: Measured clockwise from North (subtract θ from 90° if θ is angle with x -axis).
- [Diagram: Resultant vector $\mathbf{R} = 4\mathbf{i} + 3\mathbf{j}$ shown as a right-angled triangle with angle θ].

4.3 Forces and acceleration

- **Definition / Core Concept:** A non-zero resultant force causes a proportional acceleration.
- **Governing Law / Principle:**
 - **Newton's second law of motion:** The force needed to accelerate a particle is equal to the product of the mass of the particle and the acceleration produced.
 - **Gravity:** The force of attraction between an object and the Earth.
- **Conceptual and Mechanistic Explanation:**
 - A force of 1 N accelerates a mass of 1 kg at a rate of 1 m s^{-2} .
 - Weight is the force acting vertically downward due to gravity.
- **Assumptions and Conditions:**

- Acceleration due to gravity $g = 9.8 \text{ m s}^{-2}$ unless stated otherwise.
 - Earth's gravitational field is assumed uniform.
- Equations and Derivations:**
 - Equation of motion: $F = ma$.
 - Weight formula: $W = mg$.
 - Vertical equilibrium (when not moving vertically): $R - mg = 0$.
- Practical Significance:** Relates the net force applied to the resulting change in motion.
- [Diagram: Particle of mass m falling freely; labels: Weight $W = mg$ acting downward, acceleration g downward].

4.4 Motion in two dimensions

- Definition / Core Concept:** Application of $\mathbf{F} = m\mathbf{a}$ using 2D vector notation for forces and acceleration.
- Governing Principle:** Acceleration is a vector quantity; $\mathbf{F}$ and $\mathbf{a}$ are in the same direction.
- Conceptual and Mechanistic Explanation:**
 - To find acceleration from multiple forces, add force vectors first to find resultant $\mathbf{F}$, then divide by mass m .
- Notation:** Acceleration can be written as $(p\mathbf{i} + q\mathbf{j}) \text{ m s}^{-2}$ or as a column vector $\begin{pmatrix} p \\ q \end{pmatrix} \text{ m s}^{-2}$.
- Equations:**
 - Vector Second Law: $\mathbf{F} = m\mathbf{a}$.
 - Resultant Acceleration: $\mathbf{a} = \frac{\sum \mathbf{F}}{m}$.
- [Diagram: Acceleration vector $\mathbf{a}$ with components $6\mathbf{i}$ and $16\mathbf{j}$ and resultant magnitude $|\mathbf{a}| = \sqrt{6^2 + 16^2}$].

4.5 Connected particles

- Definition / Core Concept:** Systems involving motion of more than one particle.
- Governing Law / Principle:**
 - Newton's third law of motion:** For every action there is an equal and opposite reaction.
- Conceptual and Mechanistic Explanation:**
 - Particles can be considered separately (to find internal forces like tension) or as a single system (to find acceleration) if they move in the same straight line.
 - Internal forces (Tension T) cancel out when treating the system as a whole.
- Assumptions and Conditions:**
 - Light inextensible string:**
 - Tension T is the same throughout the length.

- Acceleration a is the same for both objects.
 - Mass of the string is negligible.
- **Smooth surface:** No friction.
- **Rough surface:** Friction opposes motion.
- **Equations and Derivations:**
 - For system: $(\text{External Driving Force}) - (\text{Total Resistance}) = (\sum m)a$.
 - Individual equation for particle P : $T - \text{Resistance}_P = m_P a$.
- [Diagram: Two blocks $Q(3 \text{ kg})$ and $P(5 \text{ kg})$ connected by string on rough plane; labels: Driving force 40 N , resistances 6 N and 10 N , internal tensions T].
- [Diagram: Light scale-pan with masses A and B inside being raised vertically].

4.6 Pulleys

- **Definition / Core Concept:** Systems where particles are connected by a string passing over a fixed pulley.
 - **Governing Principle:**
 - Particles connected by a string over a pulley **cannot** be treated as a single particle because they move in different directions.
 - **Conceptual and Mechanistic Explanation:**
 - Resolve vertically for each mass separately in the direction of its acceleration.
 - Force on the pulley is the resultant of the tensions in the string segments.
 - **Assumptions and Conditions:**
 - **Smooth pulley:** Tension is the same on both sides ($T_1 = T_2$).
 - **Light inextensible string:** Magnitude of acceleration is the same for both particles.
 - **Equations and Derived Forms:**
 - Particle 1 (moving up): $T - m_1 g = m_1 a$.
 - Particle 2 (moving down): $m_2 g - T = m_2 a$.
 - Eliminating T (adding equations): $g(m_2 - m_1) = (m_1 + m_2)a$.
 - Force exerted on pulley: $2T$ downward (for vertical strings).
 - **Special Cases:**
 - **String becomes slack:** When one particle hits the ground, tension T becomes zero; the other particle moves freely under gravity.
 - [Diagram: Pulley with masses $2m$ and $3m$; labels: Tensions T upward, weights $2mg$ and $3mg$ downward, acceleration a].
 - [Diagram: Particle A on horizontal table connected to hanging particle B via pulley at table edge].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Newton's Second Law:** $F = ma$ (where F is resultant force)
 - **Vector Form:** $\mathbf{F} = m\mathbf{a}$
 - **Weight:** $W = mg$
 - **Gravity Constant:** $g = 9.8 \text{ m s}^{-2}$
 - **Force Unit:** $1 \text{ N} = 1 \text{ kg m s}^{-2}$
 - **Vector Magnitude:** $|\mathbf{R}| = \sqrt{x^2 + y^2}$
 - **Equilibrium Condition:** $\sum \mathbf{F} = \mathbf{0}$ (Resultant force is zero)
 - **Newton's Third Law:** $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$ (Equal and opposite reaction)
 - **Pulley Force:** Force on pulley $= 2T$ (if strings are parallel)
 - **Connected System Acceleration:** $a = \frac{\text{Net External Force}}{\text{Total Mass}}$
-

ULTRA-CONDENSED FINAL SUMMARY

- **Statics:** Newton's 1st Law (Net $F = 0$) means equilibrium (rest or constant velocity).
- **Dynamics:** Newton's 2nd Law ($F = ma$). Acceleration is always in the direction of the net force.
- **Vectors:** Add component-wise ($\mathbf{i}, \mathbf{j}$). Resultant magnitude uses Pythagoras; direction uses trigonometry.
- **Gravity:** $W = mg$. Use $g = 9.8$ unless specified.
- **Connections:** Newton's 3rd Law (action/reaction). Systems with inextensible strings share the same magnitude of acceleration (a) and tension (T) throughout.
- **Pulleys:** Resolve each mass individually. Smooth/Light assumptions ensure constant T and a . If a mass hits the floor, the string goes slack ($T = 0$) and remaining particles move under gravity.

CHAPTER 5: FORCES AND FRICTION

5.1 Resolving forces

- **Definition / Core Concept:**
 - **Resolving a force:** Finding the component of a force that acts in a specific direction.
 - **Component:** The effect of a force in a particular direction, represented by the length of the projection onto that direction.
- **Governing Law / Principle:**
 - The component of a force of magnitude F in a direction at angle θ to the force is $F \cos \theta$.
 - **Triangle Law of Vector Addition:** Used to find the resultant of two forces acting at an angle without resolving into components.
- **Conceptual and Mechanistic Explanation:**
 - If a force F is applied at an angle to the direction of motion, only the component in that direction contributes to the motion.
 - For a right-angled triangle, $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$; thus, the component is $F \times \cos \theta$.
 - **Triangle of Forces:** If three forces act on a particle in equilibrium, they will form a closed triangle.
- **Assumptions and Conditions:**
 - Vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors in the x and y directions respectively.
 - Angles can be measured clockwise or anticlockwise; $\cos \theta = \cos(360^\circ - \theta)$.
- **Special Cases and Boundary Scenarios:**
 - **Force acts in direction D :** Component is $F \cos 0^\circ = F \times 1 = F$.
 - **Force acts at right angle to D :** Component is $F \cos 90^\circ = F \times 0 = 0$.
 - **Force acts in opposite direction to D :** Component is $F \cos 180^\circ = F \times -1 = -F$.
- **Equations and Derivations:**
 - **Component form:** $\text{Force} = p\mathbf{i} + q\mathbf{j}$.
 - **Resultant Magnitude (Cosine Rule):** $R^2 = P^2 + Q^2 - 2PQ \cos \theta$.
 - **Resultant Direction (Sine Rule):** $\frac{\sin \theta}{P} = \frac{\sin \phi}{R}$.
 - **Equilibrium:** Resolving horizontally ($R(\rightarrow)$) and vertically ($R(\uparrow)$) and equating the sum of components to zero.
- **Practical Significance:** Allows the calculation of net force and acceleration when multiple forces act at various angles on a single particle.
- [Diagram: Book being dragged along a table by force F at angle θ to the horizontal. Labels: A (start), B (projection), $AB = F \cos \theta$, and a vertical component FN].
- [Diagram: Vector addition showing forces P and Q forming a triangle with resultant R and angle θ].

5.2 Inclined planes

- **Definition / Core Concept:** Situations where an object moves or rests on a sloped surface (inclined plane).
- **Governing Law / Principle:** Forces are resolved **parallel to** and **at right angles to** the plane.
- **Conceptual and Mechanistic Explanation:**
 - The **Normal Reaction (R)** force acts at right angles to the plane, not vertically.
 - The weight of the object (mg) acts vertically downward.
 - Weight is resolved into two components relative to a plane inclined at angle α :
 - **Component down the slope:** $mg \sin \alpha$.
 - **Component perpendicular to the slope:** $mg \cos \alpha$.
- **Assumptions and Conditions:**
 - The plane is usually assumed to be smooth unless a coefficient of friction is provided.
 - Acceleration is directed along the plane.
- **Special Cases:**
 - **Equilibrium perpendicular to plane:** $R = mg \cos \alpha$ (plus/minus components of other applied forces).
 - **Acceleration down the slope:** Resultant force $F = ma$ where F is the sum of components parallel to the slope.
- **Equations:**
 - **Resolution perpendicular to plane:** $R(\perp)$.
 - **Resolution parallel to plane:** $R(//)$.
 - **Example (Slide down slope 15°):** $R = 10g \cos 15^\circ$; $10g \cos 75^\circ = 10a$.
- **Practical Significance:** Models real-world scenarios like vehicles on hills, skiers, or objects on ramps.
- [Diagram: Block of mass m on a plane inclined at 15° . Labels: R perpendicular to plane, mg vertically down, angle 15° between mg and the normal to the plane].

5.3 Friction

- **Definition / Core Concept:**
 1. **Friction:** A force that opposes motion between two **rough surfaces**.
 2. **Tendency to move:** Occurs when a force is applied but the object remains stationary due to friction.
 3. **Limiting Equilibrium:** The state where a particle is **on the point of moving**; friction has reached its maximum value.
- **Governing Law / Principle:**
 1. **Friction Inequality:** $F \leq \mu R$.
 2. **Limiting Friction Formula:** $F_{MAX} = \mu R$, where μ is the **coefficient of friction** and R is the normal reaction.

- **Conceptual and Mechanistic Explanation:**
 1. Friction increases to match the applied force until it reaches F_{MAX} .
 2. If applied force exceeds F_{MAX} , the object moves, and friction remains constant at μR .
 3. The rougher the surfaces, the higher the value of μ .
 - **Assumptions and Conditions:**
 1. **Smooth surface:** $\mu = 0$ (no friction).
 2. **Rough surface:** $\mu > 0$.
 3. Friction always acts in a direction to **oppose the motion** or the tendency of motion.
 - **Special Cases and Boundary Scenarios:**
 1. **Equilibrium:** $F < \mu R$ (not limiting) or $F = \mu R$ (limiting).
 2. **Motion:** $F = \mu R$ always, and $a > 0$.
 3. **Braking:** Friction between wheels and road determines effectiveness; wet/icy conditions reduce μ , increasing braking distance.
 - **Equations:**
 1. $F_{MAX} = \mu R$.
 2. **Equation of Motion with Friction:** $P - \mu R = ma$ (where P is driving force).
 - **Practical Significance:** Essential for understanding vehicle safety, walking, and mechanical stability.
 - [Diagram: Three stages of a block of mass m on a horizontal surface:
 1. Stationary (no horizontal force, no friction).
 2. Stationary with applied force T (Friction F opposes T , $F = T$).
 3. Moving/Limiting with applied force T (Friction $F_{MAX} = \mu R$ opposes motion)].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Component of Force:** $F \cos \theta$
- **Friction Maximum:** $F_{MAX} = \mu R$
- **Friction Inequality:** $F \leq \mu R$
- **Weight:** $W = mg$
- **Weight component (parallel to plane α):** $mg \sin \alpha$
- **Weight component (perpendicular to plane α):** $mg \cos \alpha$
- **Cosine Rule:** $a^2 = b^2 + c^2 - 2bc \cos A$
- **Sine Rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
- **Newton's Second Law:** $F = ma$

- **Coefficient of Friction Symbol:** μ (mu)
 - **Normal Reaction Symbol:** R
-

ULTRA-CONDENSED FINAL SUMMARY

- **Resolving:** Component in direction θ is $F \cos \theta$. Use $\mathbf{i}$ and $\mathbf{j}$ for 2D.
- **Inclined Planes:** Always resolve parallel (often $mg \sin \alpha$) and perpendicular (often $mg \cos \alpha$) to the slope. R is perpendicular to the surface.
- **Friction:** Opposes motion. $F \leq \mu R$.
- **Limiting Equilibrium:** Object is on the brink of moving; $F = \mu R$.
- **Dynamic Friction:** If moving, friction is always μR .
- **Methods:** Use simultaneous equations from horizontal/vertical or parallel/perpendicular resolutions. Use Sine/Cosine rules for triangle of forces in equilibrium.

CHAPTER 6: MOMENTUM AND IMPULSE

6.1 Momentum in one dimension

- **Definition / Core Concept:**
 - **Momentum:** The product of a body's mass (m) and its velocity (v).
 - **Impulse:** The product of a constant force (F) and the time (t) for which it acts.
- **Governing Law / Principle:**
 - **Impulse–momentum principle:** Impulse applied to a body is equal to its change in momentum.
 - Formula: $I = mv - mu$.
- **Conceptual and Mechanistic Explanation:**
 - Newton's second law ($F = ma$) relates force to rate of change of velocity ($a = \frac{v - u}{t}$).
 - Substituting a into $F = ma$ gives $F = m \left(\frac{v - u}{t} \right)$.
 - Rearranging gives $Ft = m(v - u) = mv - mu$, establishing that the force-time product causes the momentum change.
- **Assumptions and Conditions:**

- **Vector Nature:** Velocity, momentum, and impulse are vector quantities; a positive direction must be chosen and maintained throughout calculations.
- Mass (m) is constant during the time interval.
- **Equations and Symbols:**
 - Momentum: $\text{momentum} = mv$.
 - Impulse (I): $I = Ft$.
 - Units for Momentum/Impulse: Newton-seconds (N s) or kilogram-metres per second (kg m s^{-1}).
 - Unit Equivalence: $1 \text{ kg m s}^{-1} = (\text{kg m s}^{-2}) \text{ s} = 1 \text{ N s}$.
 - Change in momentum: $\text{final momentum} - \text{initial momentum}$.
- **Special Cases and Boundary Scenarios:**
 - **Impact with Fixed Surface:** When a particle hits a wall and rebounds, the impulse exerted *by* the wall is equal and opposite to the impulse exerted *on* the wall (Newton's 3rd Law).
 - **Mass Conversion:** Mass must be in kg ($1 \text{ tonne} = 1000 \text{ kg}$; $1 \text{ g} = 0.001 \text{ kg}$).
- **Practical Significance:** Used to calculate impact forces in sports (bat hitting ball) or mechanical events (collisions) where forces act for a short duration.
- [Diagram: Particle hitting a vertical wall. Labels: Initial velocity u towards wall, final velocity v rebounding from wall, impulse I acting away from wall].

6.2 Conservation of momentum

- **Definition / Core Concept:**
 - **Principle of Conservation of Momentum:** In a collision between two bodies, the total momentum of the system remains unchanged, provided no external forces act.
- **Governing Law / Principle:**
 - Derived from **Newton's third law:** When two bodies collide, they exert equal and opposite forces on each other for the same duration of time.
 - Consequently, the impulse on body 1 ($-I$) is equal and opposite to the impulse on body 2 ($+I$), so the net change in momentum for the whole system is zero.
- **Conceptual and Mechanistic Explanation:**
 - Sum of momenta before impact = Sum of momenta after impact.
 - The changes in individual momenta cancel each other out.
- **Assumptions and Conditions:**
 - No external horizontal forces (like friction) are considered during the impact interval.
 - The collision is modeled as instantaneous or occurring over a very short period.
- **Equations and Derivations:**
 - For two masses m_1, m_2 with initial velocities u_1, u_2 and final velocities v_1, v_2 :
 - $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$.
- **Special Cases and Boundary Scenarios:**

- **Coalescence:** If two particles join together after impact, they move with a common velocity (v). Formula becomes: $m_1u_1 + m_2u_2 = (m_1 + m_2)v$.
 - **Reversal of Direction:** If a particle's final velocity v is negative, it has reversed its direction of motion.
 - **Strings Going Taut:** When a string connecting two particles becomes taut, a "jerk" (impulse) is transmitted through the string. The conservation of momentum applies to the system at the instant the string goes taut to find the common velocity.
 - **Practical Significance:** Fundamental for analyzing vehicle crashes, billiard ball collisions, and explosive separations.
 - [Diagram: Two particles m_1 and m_2 before and after collision. Labels: "Before collision" with velocities u_1, u_2 ; "After collision" with velocities v_1, v_2 ; Impulse arrows I pointing in opposite directions between particles].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Momentum:** mv
 - **Impulse (Force/Time):** $I = Ft$
 - **Impulse–Momentum Principle:** $I = mv - mu$
 - **Conservation of Momentum:** $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
 - **Acceleration:** $a = \frac{v - u}{t}$
 - **Newton's Second Law:** $F = ma$
 - **Impulse from Newton's Second Law:** $Ft = m(v - u)$
 - **Common Velocity (Joined Particles):** $v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$
 - **Mass Symbol:** m (kilograms, kg)
 - **Velocity Symbols:** u (initial), v (final) (metres per second, m s^{-1})
 - **Impulse Symbol:** I (Newton-seconds, N s)
 - **Force Symbol:** F (Newtons, N)
 - **Time Symbol:** t (seconds, s)
-

ULTRA-CONDENSED FINAL SUMMARY

- **Momentum (p):** Vector defined as $mass \times velocity$. Units: N s or kg m s^{-1} .
- **Impulse (I):** Product of constant force and time (Ft). Equal to change in momentum ($mv - mu$).
- **Conservation Law:** Total momentum is conserved in collisions ($\sum mv_{\text{initial}} = \sum mv_{\text{final}}$). Derived from Newton's 3rd Law.
- **Vector Handling:** Direction is critical. If right is positive, left-moving objects must have negative velocity values.
- **Strings/Connections:** When a slack string becomes taut, momentum conservation determines the common speed of the connected particles.
- **Coalescence:** Joined particles behave as a single mass with a single final velocity.

CHAPTER 7: STATICS OF A PARTICLE

7.1 Static particles

- **Definition / Core Concept:** A particle or rigid body is in **static equilibrium** if it is at rest and the **resultant force** acting on it is **zero**.
- **Governing Law / Principle:**
 - To solve problems in statics, the sum of the components of all forces in any direction must be **zero**.
 - **Newton's First Law:** An object at rest remains at rest if the resultant force is zero.
- **Conceptual and Mechanistic Explanation:**
 - Forces are resolved into two perpendicular directions (usually horizontal and vertical, or parallel and perpendicular to an inclined plane).
 - The algebraic sum of the components in each direction is equated to zero to form equations.
 - **Triangle of Forces:** If three forces act on a particle in equilibrium, they will form a **closed triangle** when added vectorially; the resultant is zero because the "tip" of the last vector meets the "tail" of the first.
- **Assumptions, Conditions, and Constraints:**
 - Acceleration $a = 0 \text{ m s}^{-2}$.
 - For a rigid body in static equilibrium, the resultant moment about any point is also zero.
- **Special Cases and Boundary Scenarios:**

- **Lami's Theorem:** For three coplanar forces **A**, **B**, and **C** acting on a particle in equilibrium, the ratio of each force to the sine of the angle between the other two forces is constant.

$$\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma}$$
- Formula: $\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma}$ where α is the angle between **B** and **C**, etc.
- **Equations and Derived Forms:**
 - **Resolution:** $R(\rightarrow) = 0$ and $R(\uparrow) = 0$.
 - **Triangle rule (Sine Rule):** $\frac{P}{\sin \theta_1} = \frac{Q}{\sin \theta_2} = \frac{R}{\sin \theta_3}$.
 - **Triangle rule (Cosine Rule):** $R^2 = P^2 + Q^2 - 2PQ \cos \theta$.
- **Practical Significance:** Used to determine unknown magnitudes or directions of forces acting on structures or objects that must remain stationary.
- [Diagram: Three forces **P**, **Q**, and 4 N acting on a particle. Angles are 30° and 45° to the horizontal. A corresponding closed vector triangle is shown with labels 4 N, **P**, and **Q**].

7.2 Modelling with statics

- **Definition / Core Concept:** Creating mathematical representations of physical systems using weight, tension, and normal reactions to solve for equilibrium.
- **Governing Principle:** Force diagrams (free body diagrams) are used to isolate a particle and show all external forces acting on it.
- **Conceptual and Mechanistic Explanation:**
 - **Weight:** Acts vertically downward through the center of mass; $W = mg$.
 - **Tension:** Pulling force directed along a string/cable away from the object.
 - **Normal Reaction:** Perpendicular force from a surface.
- **Assumptions, Conditions, and Limitations:**
 - **Smooth bead:** A particle with a hole that moves freely; the **tension** T is the **same on either side** of the bead.
 - **Smooth pulley:** A pulley of negligible mass and no friction; the **tension** T is the **same on both sides** of the string.
 - **Inextensible string:** Does not stretch; mass is often neglected ("light").
- **Special Cases and Exceptions:**
 - **String over a smooth pulley at an edge:** Forces must be resolved separately for the hanging mass and the mass on the surface.
 - **Smooth surface:** No frictional component exists; the reaction force is purely normal.
- **Equations:**
 - **Weight component (parallel to plane α):** $mg \sin \alpha$.
 - **Weight component (perpendicular to plane α):** $mg \cos \alpha$.
- **Practical Significance:** Allows engineers to calculate necessary cable strengths or support reactions for stationary systems.

- [Diagram: Smooth bead Y on a light inextensible string attached to points X and Z . Labels: Tensions T along the string segments, Weight W N downward, and a horizontal force of 8 N].

7.3 Friction and static particles

- **Definition / Core Concept:**
 - **Friction:** A force opposing the **tendency** of surfaces to move relative to each other.
 - **Limiting Equilibrium:** The state when a body is **on the point of moving**; friction has reached its maximum value.
 - **Governing Law / Principle:**
 - **Friction Inequality:** $F \leq \mu R$.
 - **Limiting Friction Law:** $F_{MAX} = \mu R$, where μ is the coefficient of friction and R is the normal reaction.
 - **Conceptual and Mechanistic Explanation:**
 - Friction F is a reactive force; it is only as large as needed to prevent motion, up to F_{MAX} .
 - If the applied force is less than μR , the object remains in equilibrium and is **not** in limiting equilibrium.
 - Friction always acts in the **opposite direction** to the potential motion.
 - **Assumptions and Conditions:**
 - **Rough surface:** Surface has a non-zero coefficient of friction ($\mu > 0$).
 - **Coefficient of friction (μ):** Depends on the materials of the two surfaces in contact.
 - **Special Cases and Boundary Scenarios:**
 - **Equilibrium ($\sum F = 0$):**
 1. **Non-limiting:** $F < \mu R$.
 2. **Limiting:** $F = \mu R$ (body is "on the point of slipping").
 - **Acceleration ($\sum F = ma$):** If the applied force exceeds μR , the object moves and friction remains constant at μR .
 - **Equations and Derivations:**
 - Resolution on rough horizontal plane: $R(\uparrow) : R = mg$; $R(\rightarrow) : P - F = 0$.
 - Resolution on rough inclined plane (angle α): $R(\perp) : R = mg \cos \alpha$;
 $R(\parallel) : mg \sin \alpha - F = 0$.
 - **Practical Significance:** Determines the maximum load or slope angle possible before an object starts to slide.
 - [Diagram: Block of mass 8 kg on a rough horizontal plane. Labels: Weight $8g$ N down, Normal reaction R N up, pulling force P N, and friction F N opposing P].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Resultant Force at Equilibrium:** $\sum \mathbf{F} = 0$
 - **Horizontal Equilibrium:** $R(\rightarrow) = 0$
 - **Vertical Equilibrium:** $R(\uparrow) = 0$
 - **Weight Formula:** $W = mg$
 - **Gravity Constant:** $g = 9.8 \text{ m s}^{-2}$
 - **Friction Inequality:** $F \leq \mu R$
 - **Limiting Friction:** $F_{MAX} = \mu R$
 - **Lami's Theorem:** $\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma}$
 - **Sine Rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
 - **Cosine Rule:** $a^2 = b^2 + c^2 - 2bc \cos A$
 - **Normal Reaction Symbol:** R (or R_N)
 - **Coefficient of Friction Symbol:** μ
 - **Tension Symbol:** T
-

ULTRA-CONDENSED FINAL SUMMARY

- **Equilibrium Core:** Resultant force $\mathbf{F} = 0$ and Acceleration $\mathbf{a} = 0$.
- **Standard Technique:** Resolve forces into two perpendicular components and equate their sums to zero ($\sum F_x = 0, \sum F_y = 0$).
- **Three-Force Systems:** Use the **triangle of forces** or **Lami's Theorem** for rapid solving.
- **Modelling Essentials:**
 - **Beads/Pulleys:** Tension is uniform on both sides.
 - **Weight:** Always acts vertically downward (mg).
- **Friction Dynamics:**
 - Always opposes the direction of potential motion.
 - $F \leq \mu R$ (static).
 - **Limiting Equilibrium** ($F = \mu R$) means the object is on the brink of moving.
- **Inclined Planes:** Resolve parallel ($mg \sin \alpha$) and perpendicular ($mg \cos \alpha$) to the slope.

CHAPTER 8: MOMENTS

8.1 Moments

- **Definition / Core Concept:**
 - **Moment:** A measure of the **turning effect** of a force on a **rigid body**.
 - **Rigid Body:** A model of an object where **rotational effects** and the size of the object are considered, rather than treating it as a dimensionless particle.
- **Governing Law / Principle:** The moment of a force is the **product** of the magnitude of the force and the **perpendicular distance** from the axis of rotation to the line of action of the force.
- **Conceptual and Mechanistic Explanation:**
 - A force applied to a body can cause it to rotate about a pivot (axis).
 - If the distance provided is not perpendicular, **trigonometry** must be used to find the perpendicular component.
 - **Directional Requirement:** When describing a moment, the **direction of rotation** (clockwise or anticlockwise) must always be specified.
- **Assumptions, Conditions, and Constraints:**
 - **Lamina:** A two-dimensional object whose **thickness can be ignored**.
 - Force is assumed to act at a specific point on the rigid body.
- **Equations and Derived Forms:**
 - **Standard Formula:** Clockwise moment of F about $P = |F| \times d$.
 - **Trigonometric Formula:** Moment = $|F| \times d \sin \theta$.
 - **Units:** Newton metres (**N m**) or Newton centimetres (**N cm**).
- **Practical Significance:** Used by engineers to calculate the loads applied to structures like cranes or to determine the stability of mechanical systems.
- [Diagram: A force F acting at a distance d from pivot P at an angle θ . A dashed line shows the perpendicular distance $d \sin \theta$].
- [Diagram: A rectangular lamina $ABCD$ with a horizontal force acting at corner B].

8.2 Resultant moments

- **Definition / Core Concept:** The **algebraic sum** of the moments produced by each individual force acting on a body.
- **Governing Principle:** When several **coplanar forces** act on a body, the turning effect around a given point is determined by the resultant moment.
- **Conceptual and Mechanistic Explanation:**
 - **Coplanar forces:** Forces that act in the **same plane**.

- To find the resultant, a **positive direction** (e.g., clockwise) is chosen. Moments in the opposite direction (anticlockwise) are treated as negative values.
- **Equations:**
 - Resultant moment = $\sum(\text{individual moments})$.
- **Practical Significance:** Essential for determining whether a body will rotate and in which direction.
- [Diagram: A light rod with multiple vertical forces (3N , 2N , 4N) acting at various distances from a pivot P].

8.3 Equilibrium

- **Definition / Core Concept:** A state where a rigid body remains at rest with **no resultant force** and **no resultant turning effect**.
- **Governing Law / Principle:** For a rigid body in equilibrium:
 - The **resultant force** in any direction is 0 N .
 - The **resultant moment** about any point is 0 N m (Sum of clockwise moments = Sum of anticlockwise moments).
- **Conceptual and Mechanistic Explanation:**
 - Problems are simplified by choosing a pivot point to take moments about that **eliminates unknown forces** (the moment of a force acting through the pivot is zero).
- **Assumptions and Limitations:**
 - **Uniform Rod:** The weight acts at its **midpoint** (geometric centre).
 - **Non-bending:** The rod is assumed to be rigid and does not bend or buckle.
 - **Particles on Rod:** People or weights on a beam are modeled as particles whose weight acts at a specific point.
- **Equations and Derivations:**
 - **Vertical Resolution:** $\sum R_{\text{upward}} = \sum \text{Weights}_{\text{downward}}$.
 - **Moment Equilibrium:**
Total Clockwise Moments = Total Anticlockwise Moments.
- **Practical Significance:** Used to calculate reaction forces at supports (e.g., pillars of a bridge) and tensions in supporting cables.
- [Diagram: A uniform rod AB resting on two triangular supports C and D . Labels: Normal reactions R_C and R_D upward, Weight W acting at the center].

8.4 Centres of mass

- **Definition / Core Concept:** The **unique point** through which the entire weight of an object is considered to act.
- **Governing Principle:**
 - For a **uniform rod**, the centre of mass is at its **midpoint**.
 - For a **non-uniform rod**, the centre of mass is **not necessarily** at the midpoint.

- **Conceptual and Mechanistic Explanation:**
 - The position of the centre of mass can be found by taking moments about a pivot. The total weight of the rod is treated as a single force acting at distance x from the pivot.
- **Equations:**
 - $W \times \bar{x} = \sum (w_i \times x_i)$ where W is total weight and $\bar{x}$ is the distance to the centre of mass.
- **Practical Significance:** Necessary for balancing objects (e.g., a see-saw) and determining the stability of structures.
- [Diagram: A non-uniform plank AB with weight $25g \text{ N}$ acting at a point M which is not the midpoint].

8.5 Tilting

- **Definition / Core Concept:** The state where a rigid body is **on the point of rotating** about a specific pivot.
- **Governing Law / Principle:** When a rigid body is on the point of tilting about a pivot, the **reaction at any other support** (or the tension in any other supporting wire/string) is **zero (0)**.
- **Conceptual and Mechanistic Explanation:**
 - As a load moves toward one end of a supported beam, the reaction at the distant support decreases.
 - At the moment of tilting, the beam loses contact with the distant support, making the reaction force at that support vanish.
- **Conditions:**
 - Resultant moment about the tilting pivot must be zero for limiting equilibrium.
 - Reaction at the non-tilting support $R = 0$.
- **Practical Significance:** Used to determine the maximum distance a person can walk on a plank or the maximum load a crane can lift before it capsizes.
- [Diagram: A uniform beam AB on supports C and D . A child stands at end A . The beam is shown tilting about C with the beam lifted off support D].

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Moment of force (perpendicular):** $|F| \times d$
- **Moment of force (at angle):** $|F| \times d \sin \theta$
- **Moment Units:** N m (Newton metres) or N cm (Newton centimetres)

- **Equilibrium (Force):** $\sum \mathbf{F} = 0$
 - **Equilibrium (Moment):** $\sum \text{Moments} = 0$
 - **Tilting Condition (Reaction at other support):** $R = 0$
 - **Weight:** $W = mg$
 - **Normal Reaction Symbol:** R or R_N
 - **Centre of Mass Position:** $\bar{x}$
 - **Total Momentum (System):** $\sum m_i u_i = \sum m_i v_i$ (Contextual link to Chapter 6).
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ULTRA-CONDENSED FINAL SUMMARY

- **Moment definition:** Turning effect = Force $\times$ Perpendicular distance.
- **Rigid Body vs Particle:** Rigid bodies have size/rotation; particles are points.
- **Equilibrium requirements:**
 1. Net Force = 0.
 2. Net Moment about **any** point = 0.
- **Uniform vs Non-uniform:** Uniform objects have the centre of mass at the midpoint; non-uniform objects require moment calculations to locate it.
- **Tilting Rule:** If an object is about to tilt about pivot A , the reaction at support B is **zero**.
- **Problem-solving tip:** Take moments about a point with the most unknown forces to simplify equations. Always state the direction (Clockwise/Anticlockwise).